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Universidade do Minho
Escola de Engenharia
Departamento de Informática

Context Engineering & Modern RAG

MESTRADO EM INTELIGÊNCIA ARTIFICIAL

Inteligência Artificial Generativa

2026/2027

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Agenda



Introduction	Context engineering	Modern RAG	Live quiz	summary
Why did the model get it wrong?	Anatomy, failure modes, write · select · compress · isolate	Chunking, embeddings, hybrid search, reranking, GraphRAG, agentic RAG	5 questions, teams	

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Introduction: why was it wrong?

The scenario

A company chatbot is asked: ***"What is our refund window for annual plans?"***

It confidently answers "14 days". The real policy, updated last month, is 30 days.

A The model was never trained on the policy

B The old policy was retrieved, not the new one

C The right document was there, but buried mid-context

D The system prompt told it to always answer

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Part 1 · Context engineering

Curating the smallest set of high-signal tokens

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Prompt engineering → context engineering

Prompt engineering

- Craft one instruction well
- Single turn, static text
- Wording, role, few-shot examples
- *“What should I say?”*

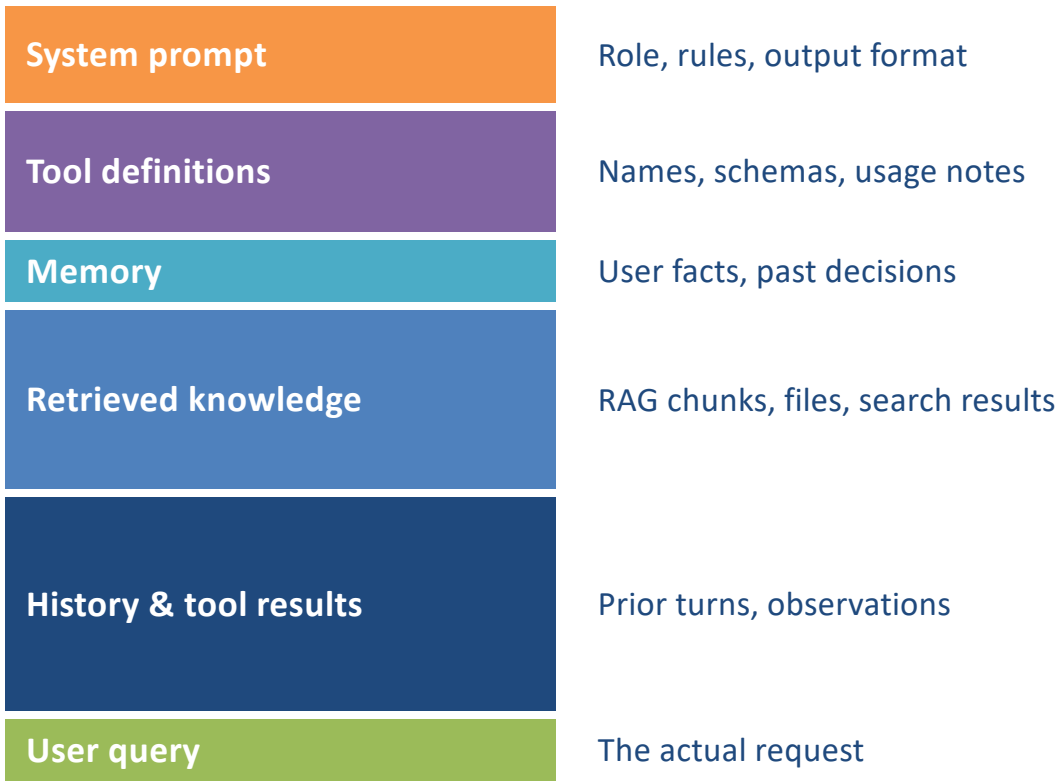
Context engineering

- Assemble the whole window, every turn
- Dynamic: tools, memory, retrieval, history
- Budgeting, ordering, pruning, caching
- *“What does the model need to see now?”*

2025, Karpathy, Tobi Lütke, blog da Anthropic "Effective context engineering for AI agents", LangChain

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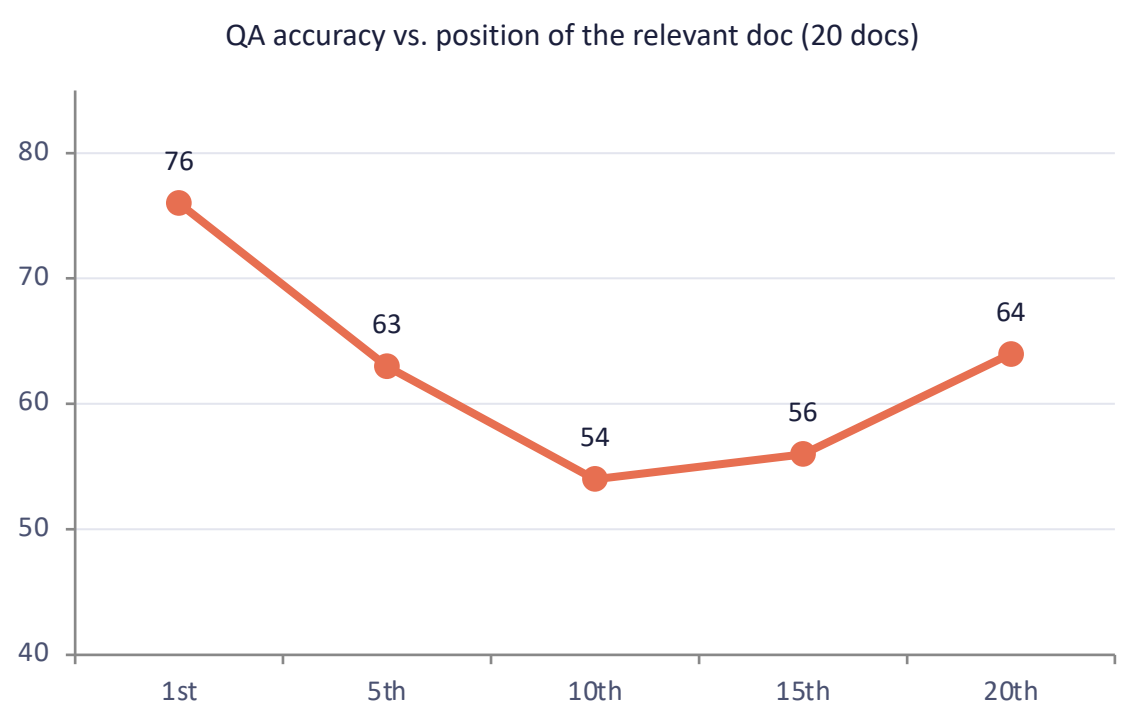
Anatomy of a context window



Every token competes for attention

- Attention is a finite budget: more tokens ≠ better answers.
- The user query is often under 1% of the window in agentic systems.
- Stable parts go first (cacheable); volatile parts go last.

Long context is not free: lost in the middle



Illustrative U-shape, after Liu et al. (2023), "Lost in the Middle"

Context rot

Recall degrades as the window fills, even within the advertised limit.

Position matters

Put key docs at the start or end; the query near the end.

Less is more

Five relevant chunks often beat fifty loosely related ones.

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Four ways context fails



Poisoning

A hallucination or error enters the context and is reused as fact in later steps.



Distraction

The context grows so long the model over-focuses on it and neglects what it learned.



Confusion

Superfluous content (e.g. 40 irrelevant tools) degrades the response.



Clash

Parts of the context contradict each other: old vs. new policy, conflicting tool outputs.

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Four strategies: write, select, compress, isolate



Write

Persist info outside the window

Scratchpads, notes files, long-term memory



Select

Pull in only what is relevant

RAG, memory retrieval, tool selection



Compress

Keep the signal, drop tokens

Summaries, compaction, trimming tool output



Isolate

Split context across units

Sub-agents, sandboxes, state objects

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Structuring what goes in

```
<system>
  You are a support agent for Acme.
  If unsure, say so. Cite doc ids.
</system>
<documents>
  <doc id="v3" date="2026-09">…</doc>
</documents>
<examples> …2-3 canonical cases…
</examples>
<question>
  What is the refund window?
</question>
```

Delimit with tags

Clear sections stop documents being read as instructions.

Add metadata

Ids, dates and sources let the model resolve clashes and cite.

Right altitude

Principles over brittle if-else rules; examples over long lists.

Long docs first

Material at the top, question at the end.

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Memory: what persists beyond one window

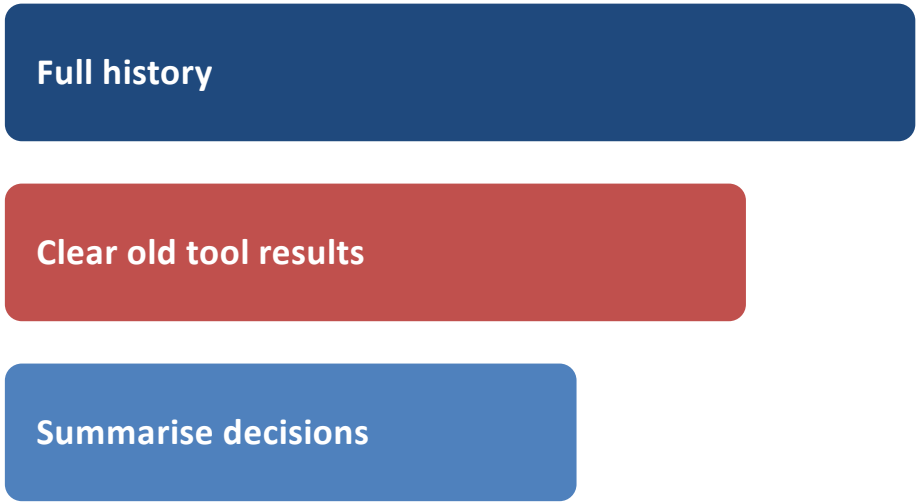
Working	The current context window: instructions, recent turns, scratchpad.
Episodic	Past interactions and outcomes, e.g. “last time this fix failed”.
Semantic	Stable facts about user or domain, e.g. “prefers Python”.
Procedural	How to do things: rules, skills, system prompts that evolve.

Design questions

- What gets written, and who decides?
- How is it retrieved: recency, similarity, importance?
- How is stale memory updated or forgotten?
- Can the user see and delete it?

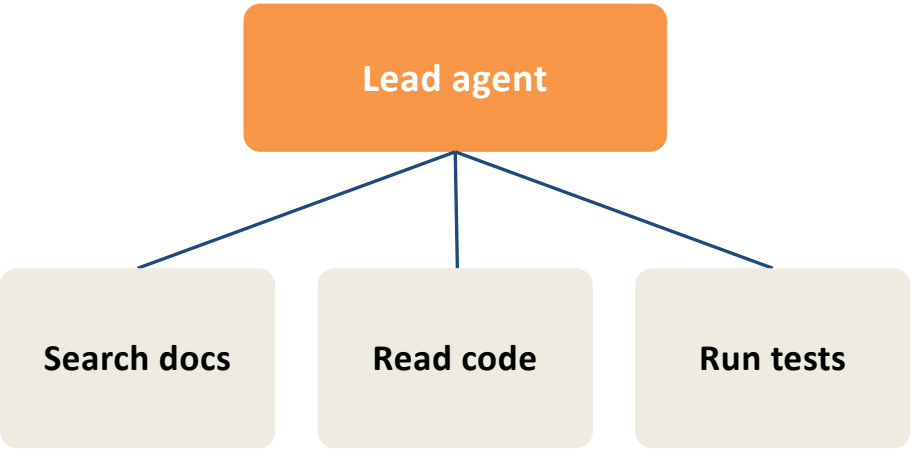
Compress and isolate in long-running agents

Compaction



Restart the window with the summary plus the most recent files.

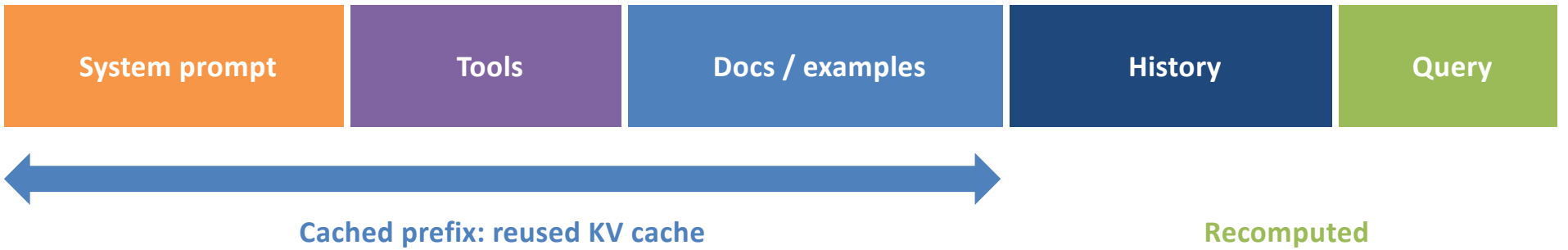
Isolation with sub-agents



Each sub-agent explores with a clean window and returns a short summary, not its full trace.

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Order for the cache: stable prefix, volatile suffix



Cheaper & faster

Providers bill cached input tokens at a fraction of the normal price and cut time-to-first-token.

One change breaks it

A timestamp at the top of the system prompt invalidates everything after it.

Append, don't edit

Keep history append-only; avoid reshuffling tools mid-session.

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Activity 1 · Context surgery

A support agent's window is at 180k tokens and answers are getting worse. It contains:

- 4k system prompt with today's date on line 1
- 52 tool definitions (the agent uses 6)
- 3 versions of the refund policy (2023, 2025, 2026)
- 40 raw log dumps from earlier tool calls
- Full 120-turn chat history
- The user's new question

In pairs

1. Label each item: keep, write, select, compress or isolate.
2. Reorder what remains for caching.
3. Estimate your new token budget.

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Part 2 · Modern RAG

Select: getting the right tokens into the window

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RAG vs. long context vs. fine-tuning

Need	RAG	Long context	Fine-tuning
Fresh, changing knowledge	Strong: re-index	OK, per query	Weak: retrain
Citations & traceability	Strong	Partial	Weak
Cost per query	Low	High at scale	Low
Huge corpora (millions of docs)	Yes	No	No
Style, format, behaviour	Weak	Partial	Strong
Access control per user	Strong: filter	Manual	Weak

They combine: fine-tune for behaviour, retrieve for knowledge, use long context for the retrieved set.

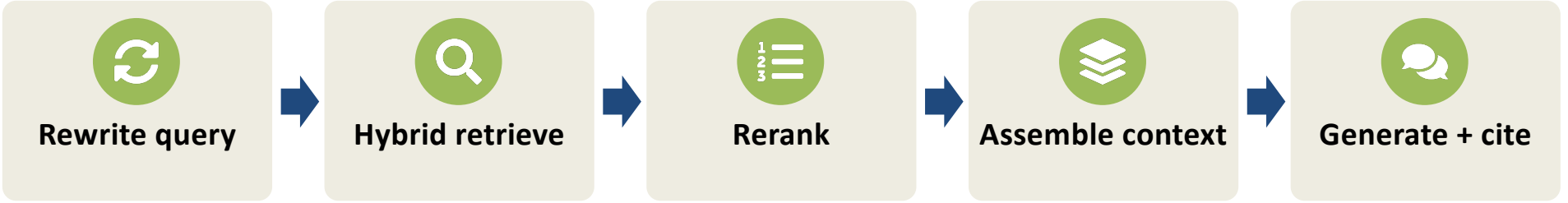
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The modern RAG pipeline

OFFLINE · indexing



ONLINE · per query



Most quality problems start upstream: bad parsing and chunking cannot be fixed by a better LLM.

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Chunking: the most underrated decision

1	**Fixed-size**	N tokens with overlap. Simple baseline; cuts mid-sentence.
2	**Recursive**	Split on paragraphs, then sentences, until size fits. Solid default.
3	**Structure-aware**	Follow headings, sections, tables, code blocks. Best for docs.
4	**Semantic**	Split where embedding similarity between sentences drops.

Fix the lost context

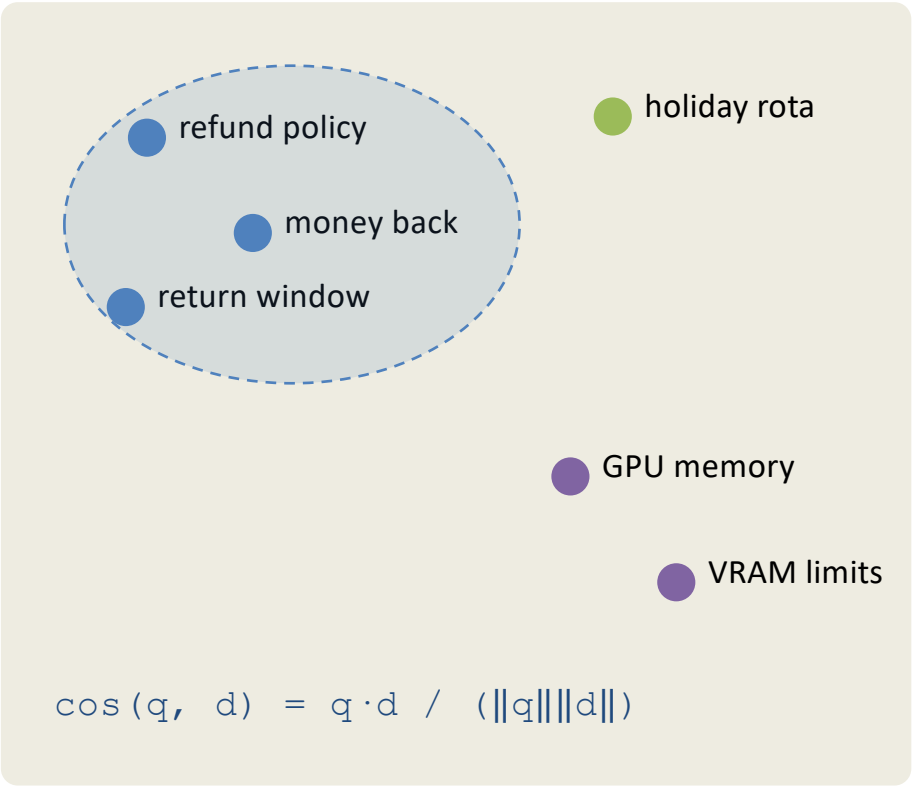
Contextual retrieval: an LLM prepends a short summary of where the chunk sits in the document before embedding.

Late chunking: embed the whole document, then pool token vectors per chunk.

Parent-child: match small chunks, return the larger parent.

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Embeddings: meaning as geometry



Bi-encoder

Query and doc encoded separately, so docs are pre-computed and search is fast.

Dense vs. sparse

Dense captures meaning; sparse (BM25, SPLADE) captures exact terms and codes.

Matryoshka

Truncate vectors (e.g. 1024 → 256 dims) with small quality loss: cheaper storage.

Choose wisely

Check MTEB/MMTEB, language (PT!), domain and context length; then test on your data.

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Vector indexes: trading recall for speed



Flat (exact)

Compare against every vector

100% recall, slow beyond ~1M



HNSW

Multi-layer proximity graph

High recall, fast, memory-hungry



IVF

Cluster, then search nearest clusters

Tunable, good for large sets



PQ / quantised

Compress vectors into codes

Low memory, some recall loss

Where they live: pgvector, Elasticsearch/OpenSearch, Qdrant, Weaviate, Milvus, FAISS, LanceDB. Pick for filtering, hybrid support and ops, not only speed.

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Hybrid search: keywords + meaning

BM25 (sparse)

Exact terms: “Art. 23”, “SKU-4471”, names

Dense vectors

Paraphrase: “money back” ≈ “refund”

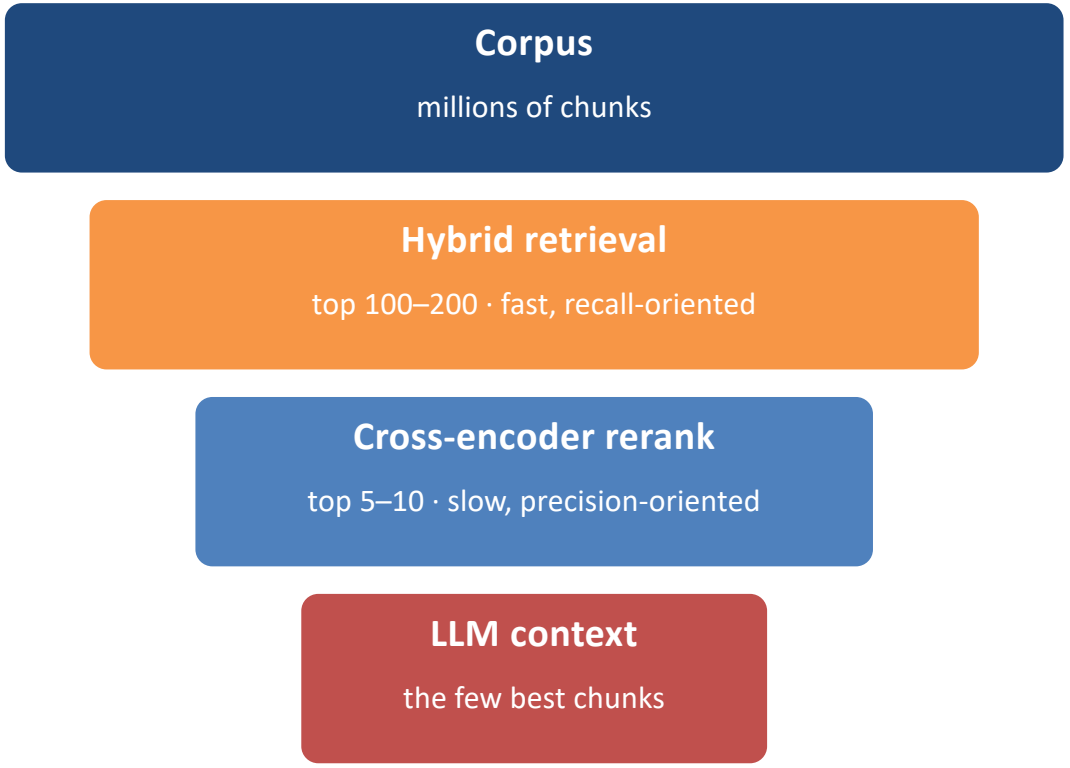
Reciprocal Rank Fusion

$$RRF(d) = \Sigma\ 1/(k + rank_i(d)),\ k=60$$

- Uses ranks, not scores: no need to normalise BM25 vs. cosine.
- A doc ranked 1st by BM25 and 3rd by dense: 1/61 + 1/63 ≈ 0.032
- Alternative: weighted score fusion, α·dense + (1−α)·sparse

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Reranking: a two-stage funnel



Bi-encoder

embed(q) · embed(d)
No interaction between query and doc tokens.

Cross-encoder

score([q ; d])
Full attention across both: far more accurate, too slow for the whole corpus.

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Fix the question before searching

Rewrite

Turn chat context into a standalone query.

“and for annual plans?” → “refund window for annual plans”

Multi-query

Several phrasings, retrieve each, fuse.

Boosts recall on vague questions

Decompose

Split multi-hop questions into parts.

“Compare 2025 and 2026 policy” → 2 searches

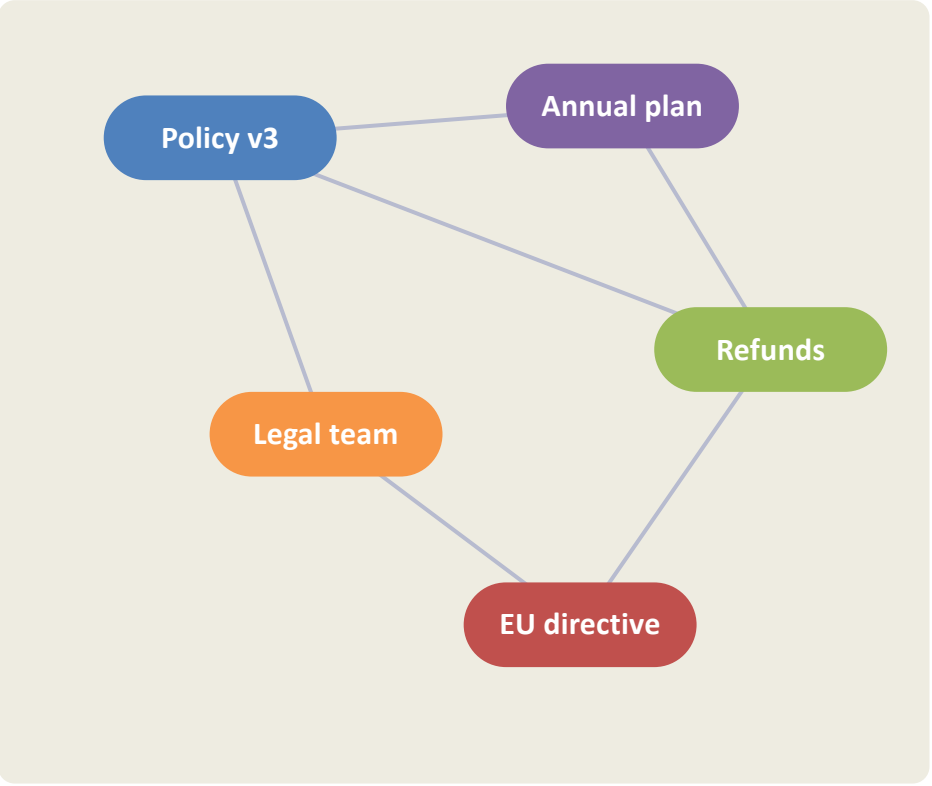
HyDE

Embed a guessed answer; search with it.

Answer-to-answer similarity

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GraphRAG: answers across the corpus



1 · Extract

LLM pulls entities and relations from every chunk into a knowledge graph.

2 · Cluster & summarise

Community detection (e.g. Leiden); an LLM writes a summary per community.

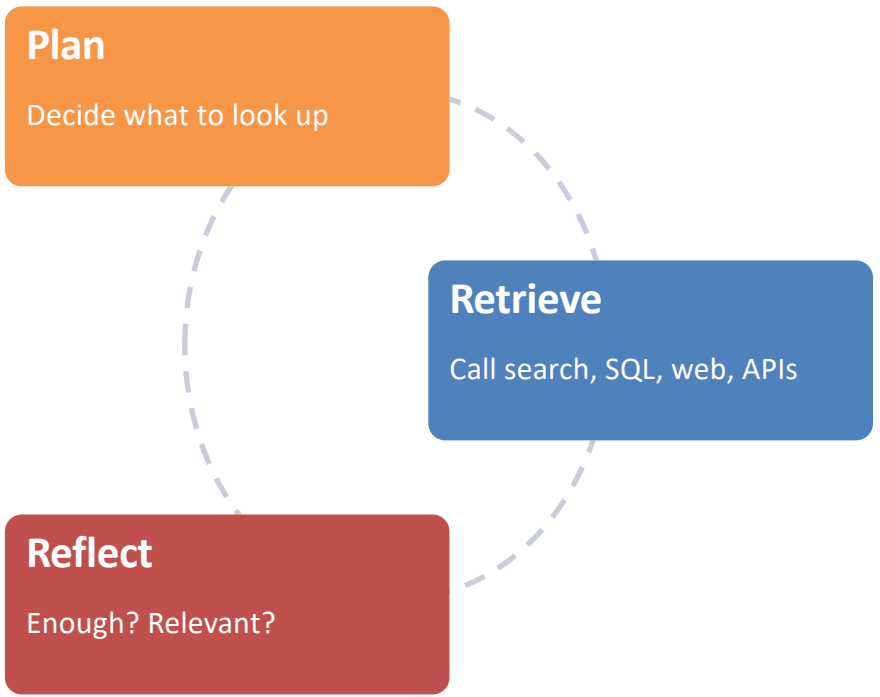
3 · Query

Local: walk from matched entities. Global: map-reduce over community summaries.

Good for “What are the main themes?”. Costly to build; overkill for simple lookups.

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Agentic RAG: retrieval becomes a tool



loop until confident, then answer

Router	Pick the source: vector DB, SQL, web, or no retrieval at all.
Self-RAG / CRAG	Grade retrieved docs; re-search or fall back to web if they are weak.
Deep research	Many search–read–reason cycles, often with sub-agents in parallel.

Trade-off: better on hard questions, but more latency, cost and failure modes.

Evaluate retrieval and generation separately

Retrieval

Recall@k: is the right chunk in the top k?

MRR: how high is the first relevant hit?

nDCG@k: graded relevance, position-weighted

Context precision: share of retrieved chunks that are useful

Generation

Faithfulness: is every claim supported by the context?

Answer relevance: does it address the question?

Citation accuracy: do sources back the sentence?

Abstention: does it say “I don’t know” when it should?

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Activity 2 · Design a RAG system

A

University assistant

Answers students on academic regulations, fees and deadlines. PT-PT, PDFs updated yearly, some tables.

B

Hospital protocols

Clinicians query 3,000 internal protocols. Must cite; wrong answers are dangerous; access by department.

C

Customer emails

Leadership asks “what are customers complaining about this quarter?” over 200k emails.

Groups of 3–4

1. Chunking strategy
2. Retrieval: dense, hybrid, graph?
3. Reranking & filters
4. Classic or agentic?
5. Two evaluation metrics
6. Biggest risk

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Live quiz · teams

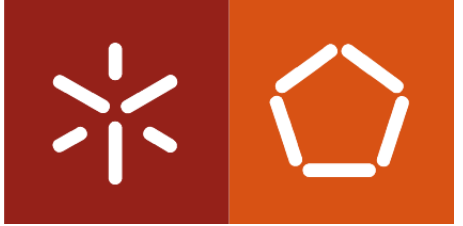
1. Why put the current date at the end of the prompt rather than the top?
2. Which failure mode: two policy versions contradict each other?
3. A user searches for “SKU-4471”. Dense-only retrieval fails. Fix?
4. Why not run the cross-encoder over the whole corpus?
5. Best approach for “main complaint themes across 200k emails”?

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Five things to remember

The window is a budget	Smallest set of high-signal tokens; more is not better.
Write · select · compress · isolate	Four levers for any agent or chatbot.
Retrieval quality starts upstream	Parsing and chunking beat model upgrades.
Hybrid + rerank is the baseline	Then add query rewriting, graphs or agents if needed.
Measure both halves	Recall@k for retrieval, faithfulness for generation.

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